

Two-State CTMC

Macroscopic-Microscopic
Equivalence in the Two-State
Continuous Time Markov Chain

Outline

1. 2-State CTMC: An Introduction

- Macroscopic Description
 - Approach: Master Equation
- Microscopic Description
 - Approach: Gillespie Algorithm

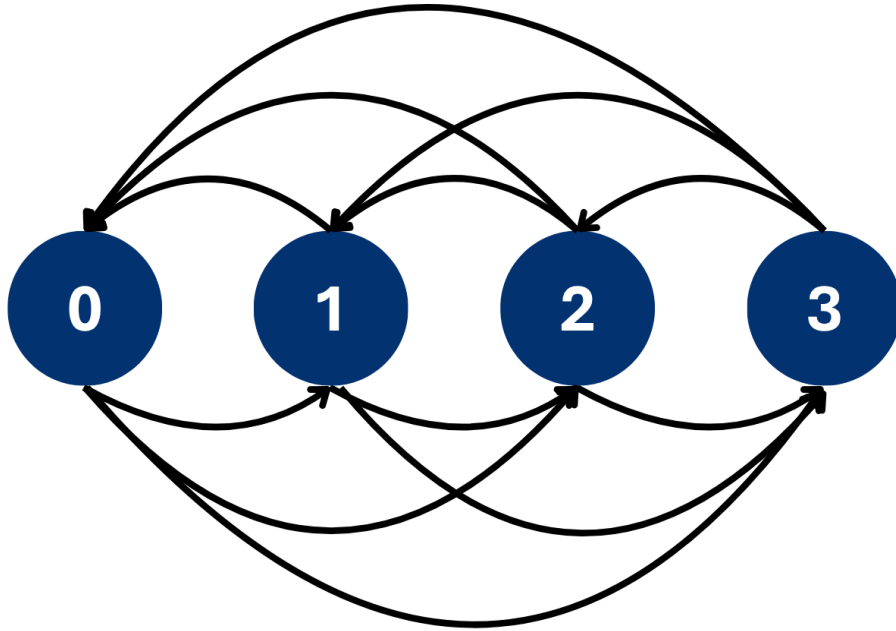
2. Statistical Analysis of the 2-State CTMC

- Equivalence between Macroscopic and Microscopic Descriptions
- Further Analyses
 - Connection to Central Limit Theorem
 - Generalization to N-state CTMC

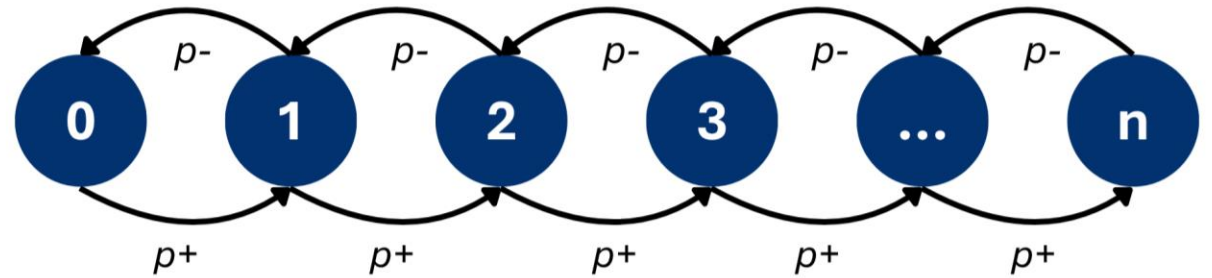
3. Conclusions and Recommendations

Markov Chain (MC)

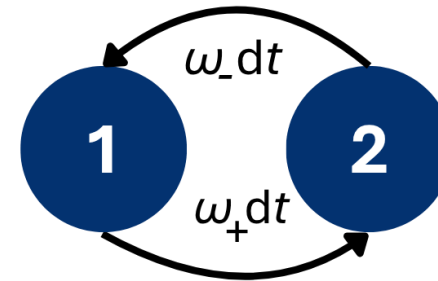
discrete-state stochastic process, conditional on the present, x_t
wherein **the future** ($x_{t+1}, x_{t+2}, \dots$) **is independent of the past** ($x_{t-1}, x_{t-2}, \dots$)



4-State MC



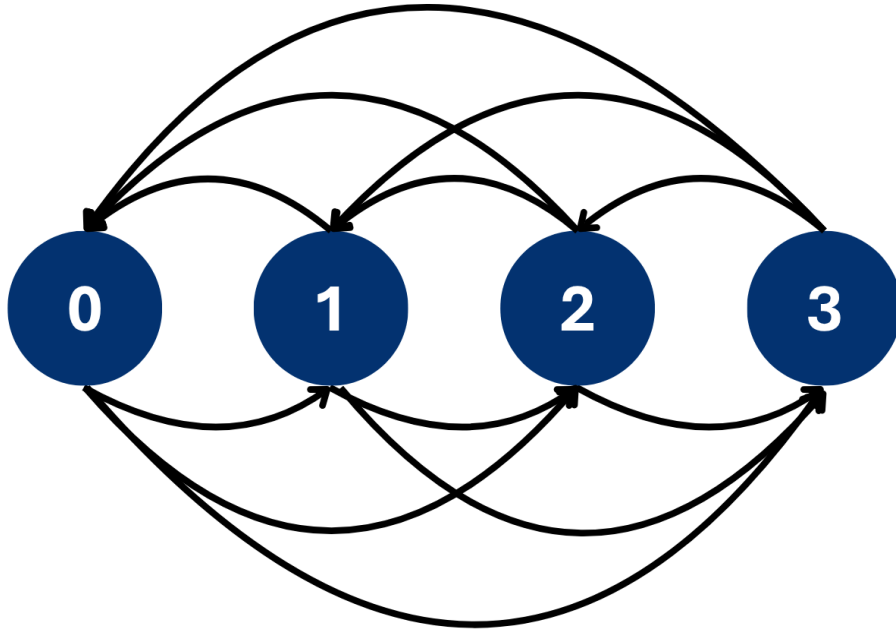
Random Walk



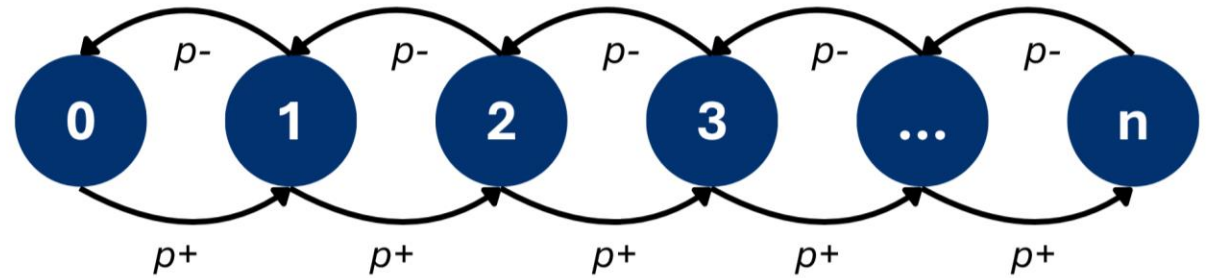
2-State MC

Markov Chain (MC)

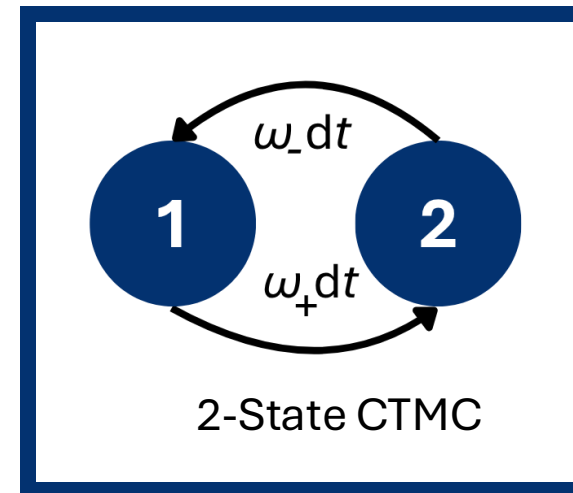
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4-State MC

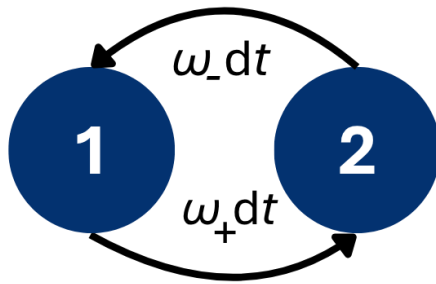


Random Walk



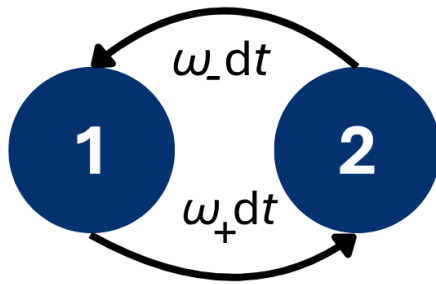
2-State CTMC

2-State CTMC: Macroscopic Description



$$\frac{dp_1}{dt} = -\omega_+ p_1 + \omega_- p_2$$
$$\frac{dp_2}{dt} = \omega_+ p_1 - \omega_- p_2$$

2-State CTMC: Macroscopic Description



$$\frac{d\vec{p}}{dt} = \mathbb{L}\vec{p}$$

$$\mathbb{L} = \begin{bmatrix} -\omega_+ & \omega_- \\ \omega_+ & -\omega_- \end{bmatrix}$$

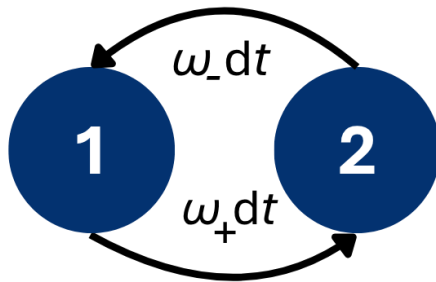
rate matrix

$$\vec{p}(t) = [p_1(t), p_2(t)]^T$$

occupation probability

$$\frac{d\vec{p}(t)}{dt} = \left[\frac{dp_1(t)}{dt}, \frac{dp_2(t)}{dt} \right]^T$$

2-State CTMC: Macroscopic Description



$$\vec{p} = \exp(\mathbb{L}t)\vec{p}(0)$$

$$\vec{p}(0) = [\delta_{n_0,1}, \delta_{n_0,2}]$$

$$\mathbb{L} = \begin{bmatrix} -\omega_+ & \omega_- \\ \omega_+ & -\omega_- \end{bmatrix}$$

rate matrix

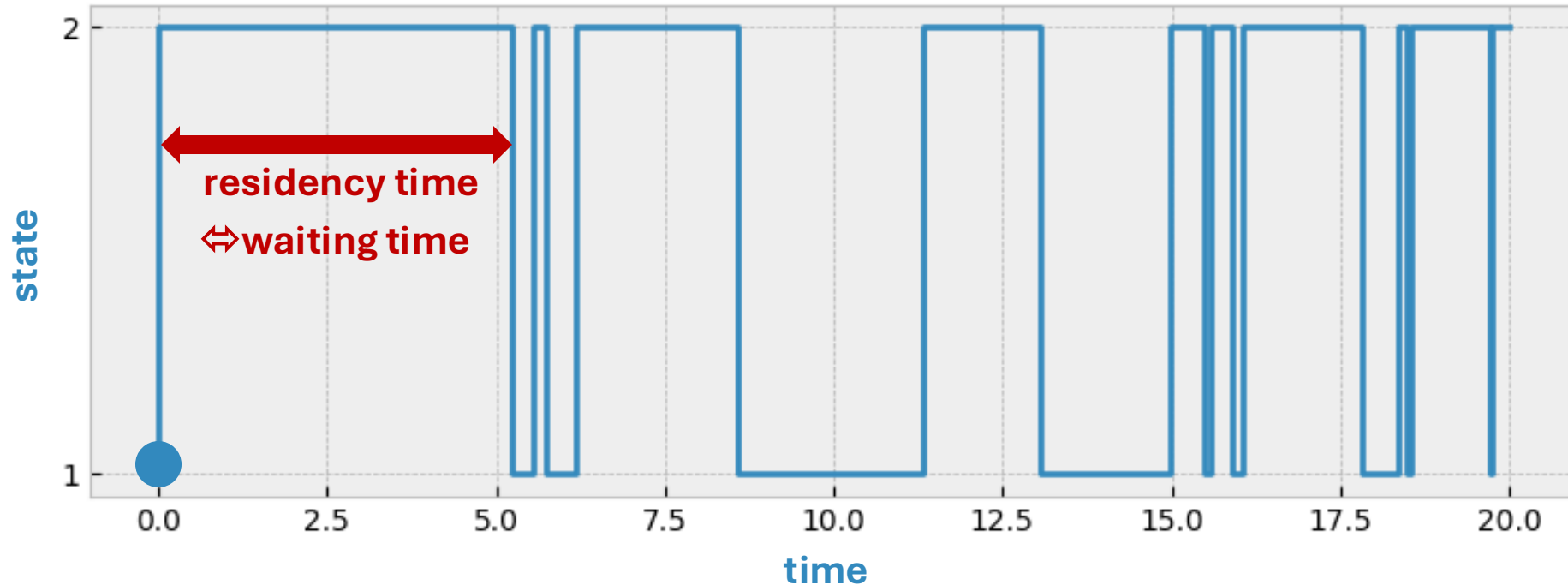
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occupation probability

$$\frac{d\vec{p}(t)}{dt} = \left[\frac{dp_1(t)}{dt}, \frac{dp_2(t)}{dt} \right]^T$$

2-State CTMC Microscopic Description

Approach: Simulate individual trajectories



Particle jumps from one state to another, with an exponentially distributed residency time.

$$p_n^{\text{empirical}}(t) = \frac{1}{\# \text{trajectories}} \sum_{\text{trajectories}} \delta(S(t) = n) \quad n \in \{1, 2\}$$

2-State CTMC Microscopic Description

Set Parameters and Generate the Trajectories

Simulate trajectories of a 2-State CTMC under specified values of **(1) transition rates**, **(2) initial states**, **(3) numbers of independent trajectories**, and **(4) maximum observation times**.

Resample for Uniformity

Convert Gillespie-generated trajectories into uniformly sampled stepwise trajectories for further concurrent analysis.

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Store Data in a Dictionary

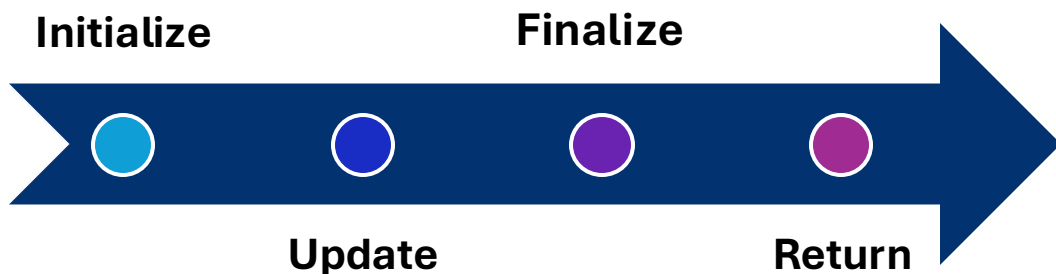
Store all trajectories in one Python dictionary, with each entry holding a nested dictionary:
`{ time: [...], position: [...], waiting time: [...] }`

Compute and Analyze Statistical Quantities

Compute statistical quantities from trajectory data, including **occupation probabilities** and **instantaneous average state currents**.

2-State CTMC Microscopic Description

Set Parameters and Generate the Trajectories



Inputs

1. $t_{\max}$ — maximum simulation time
2. ω_+ — rate of transition 1 $\rightarrow$ 2
3. ω_- — rate of transition 2 $\rightarrow$ 1
4. n_0 — initial state (1 or 2)

Outputs

1. t — array of sampling times plus final $t_{\max}$
 n — array of states at each sampling time
2. τ — array of exponentially distributed waiting times between jumps

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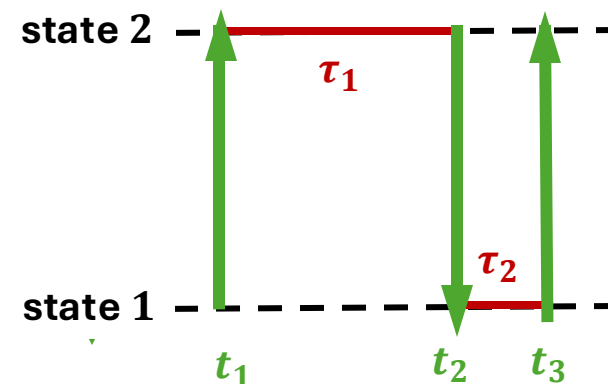
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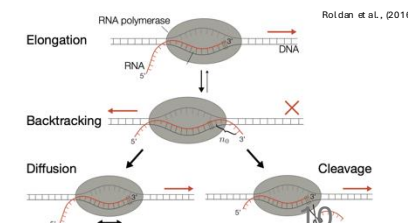
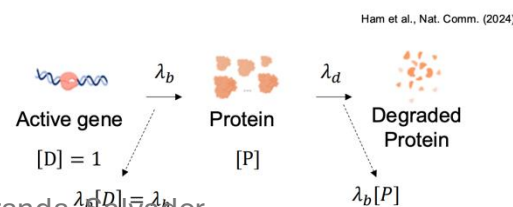
GILLESPIE ALGORITHM

simulation of waiting times, τ between 2 events



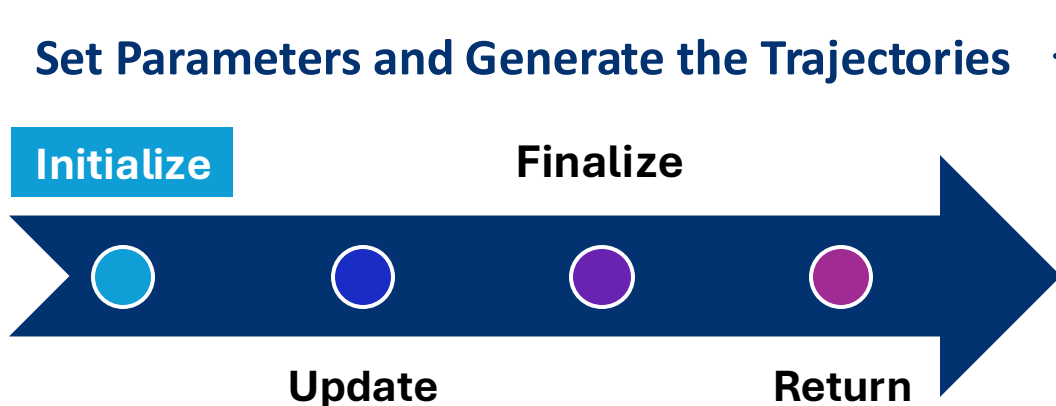
Waiting times, $\{\tau_k\}$ ($k \in \{1, 2, \dots, n\}$) are exponentially distributed with rate $= 1/\omega_i$ ($i \in \{+, -\}$)

Applications: chemical reactions, enzyme dynamics, etc.



2-State CTMC Microscopic Description

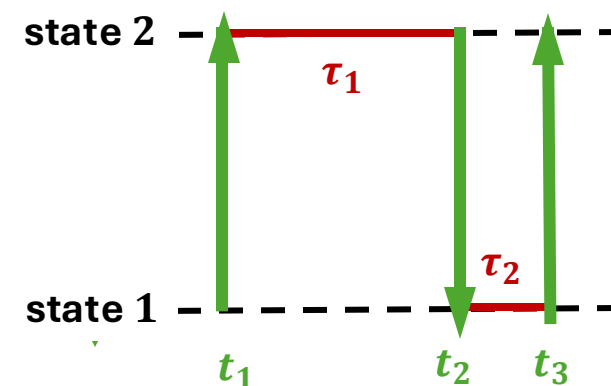
Set Parameters and Generate the Trajectories



- Set initial values, t_0, n_0 .
- Create lists to store t_k, n_k, τ_k .

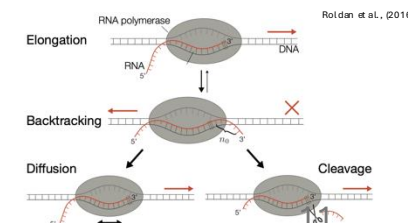
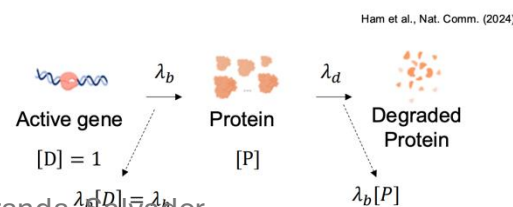
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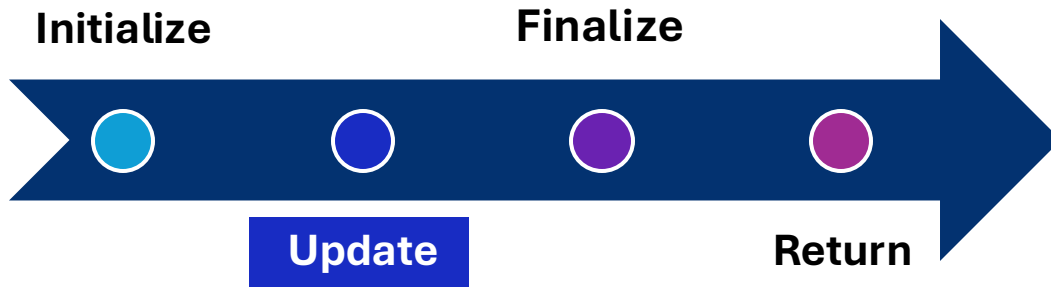
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2-State CTMC Microscopic Description

Set Parameters and Generate the Trajectories

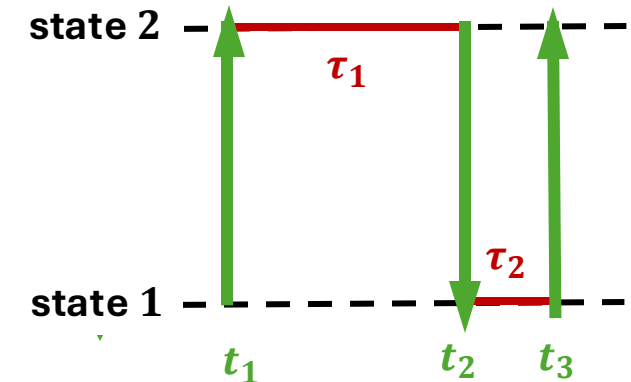


While $t_k + \tau \leq t_{max}$:

1. Sample $u \sim U(0,1)$ and transform to $r = \ln(1/u) \sim \text{Exp}(1)$
2. Compute waiting time, $\tau_{\pm} = r/\omega_{\pm}$
3. Update [timepoint], $t_{k+1} = t_k + \tau$
4. Update [position], $n_{k+1} \neq n_k$

GILLESPIE ALGORITHM

simulation of waiting times, τ between 2 events



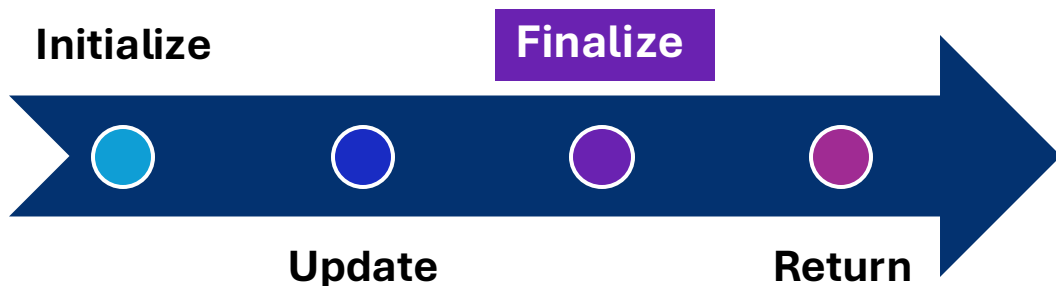
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Applications: chemical reactions, enzyme dynamics, etc.



2-State CTMC Microscopic Description

Set Parameters and Generate the Trajectories



When $t_k + \tau > t_{max}$:

1. Update [timepoint], $t_{k+1} = t_{max}$
2. Update [position], $n_{k+1} = n_k$

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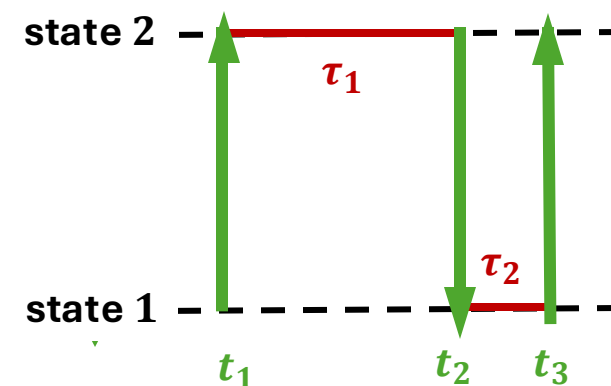
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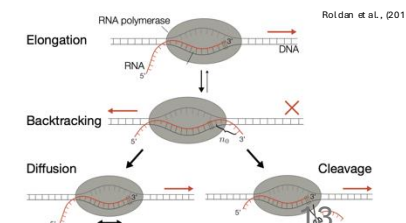
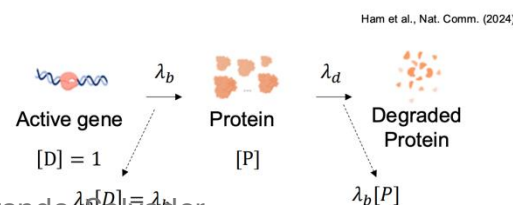
GILLESPIE ALGORITHM

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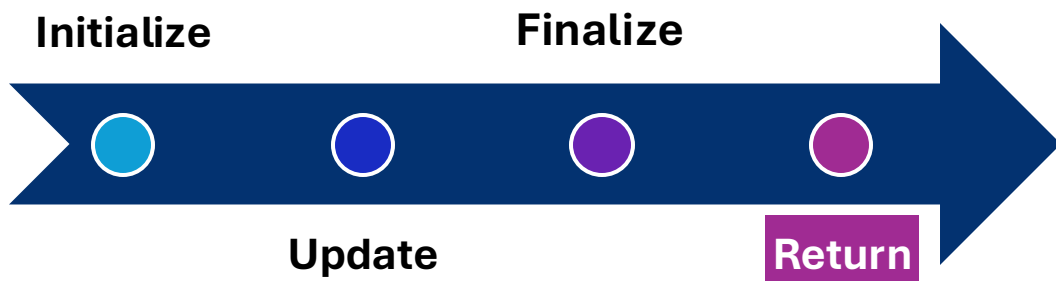
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Applications: chemical reactions, enzyme dynamics, etc.



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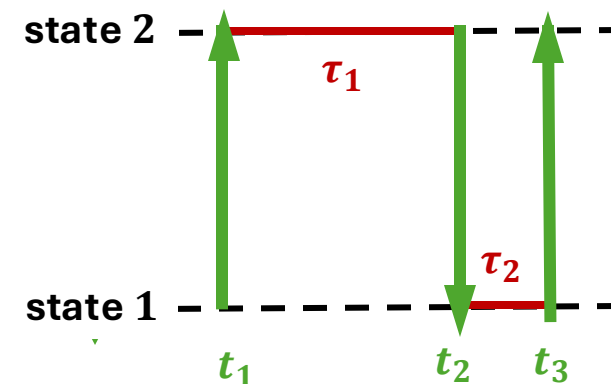
Set Parameters and Generate the Trajectories



Convert lists of t, n, τ to NumPy Arrays and return.

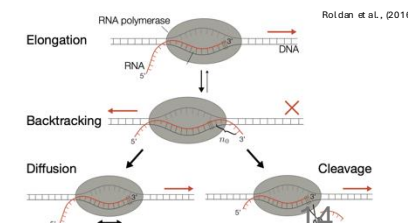
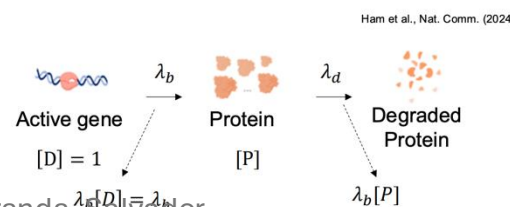
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simulation of waiting times, τ between 2 events



Waiting times, $\{\tau_k\}$ ($k \in \{1, 2, \dots, n\}$) are **exponentially distributed** with **rate** $= 1/\omega_i$ ($i \in \{+, -\}$)

Applications: chemical reactions, enzyme dynamics, etc.



2-State CTMC Microscopic Description

- All results are stored in a single Python dictionary called **trajectories**.
- Each trajectory is stored under a descriptive string key of the form "**w**+{ **ω_+** }_**w**-{ **ω_-** }_**r**{**i**}" where **i** is the trajectory index.
- Each entry of trajectories is itself a dictionary with:
 - **time** — array of recorded time points
 - **position** — array of corresponding states (1 or 2)
 - **waiting time** — array of exponentially distributed waiting times between jumps

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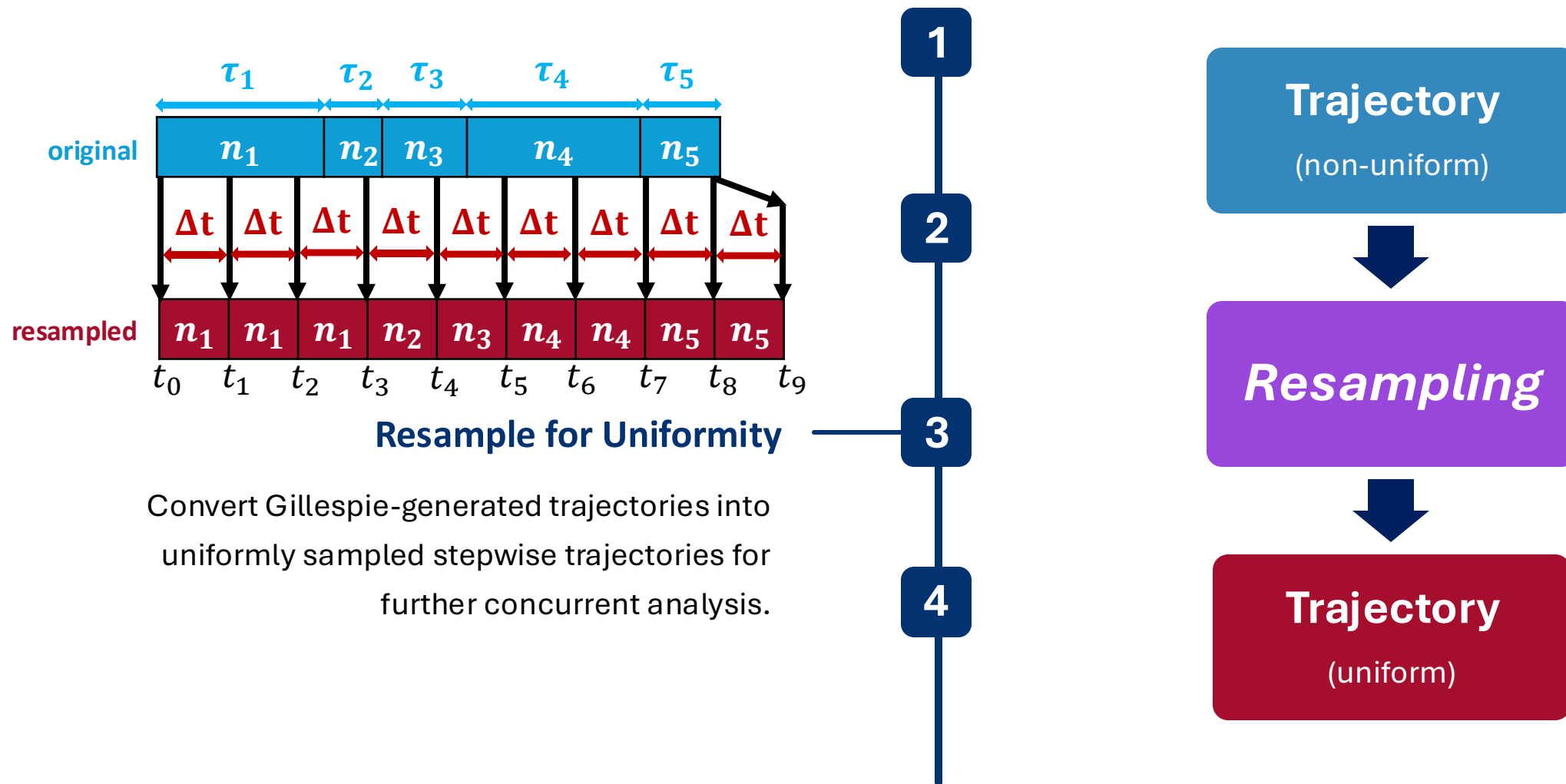
Store Data in a Dictionary

Store all trajectories in one Python dictionary, with each entry holding a nested dictionary:
{ time: [...], position: [...], waiting time: [...] }

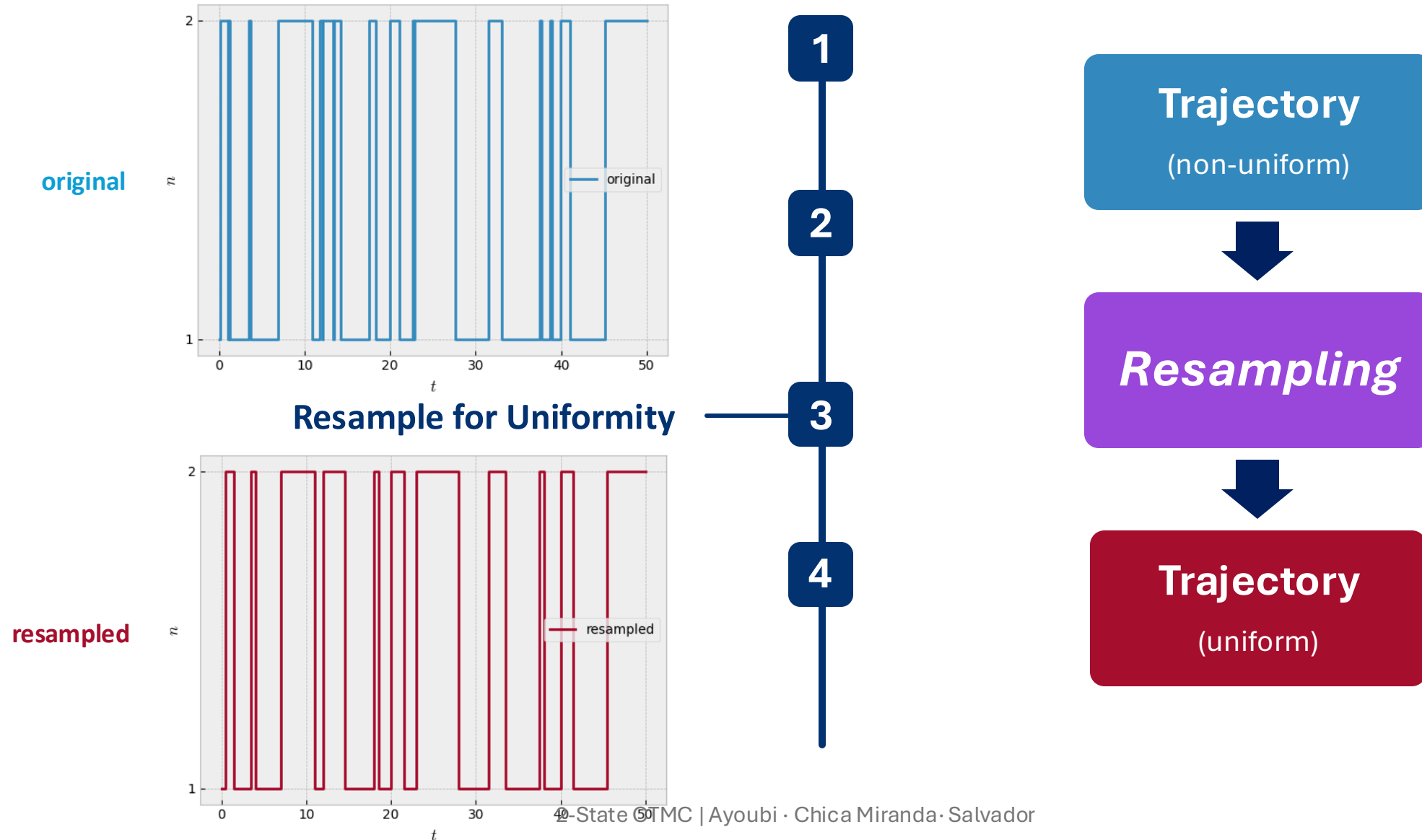
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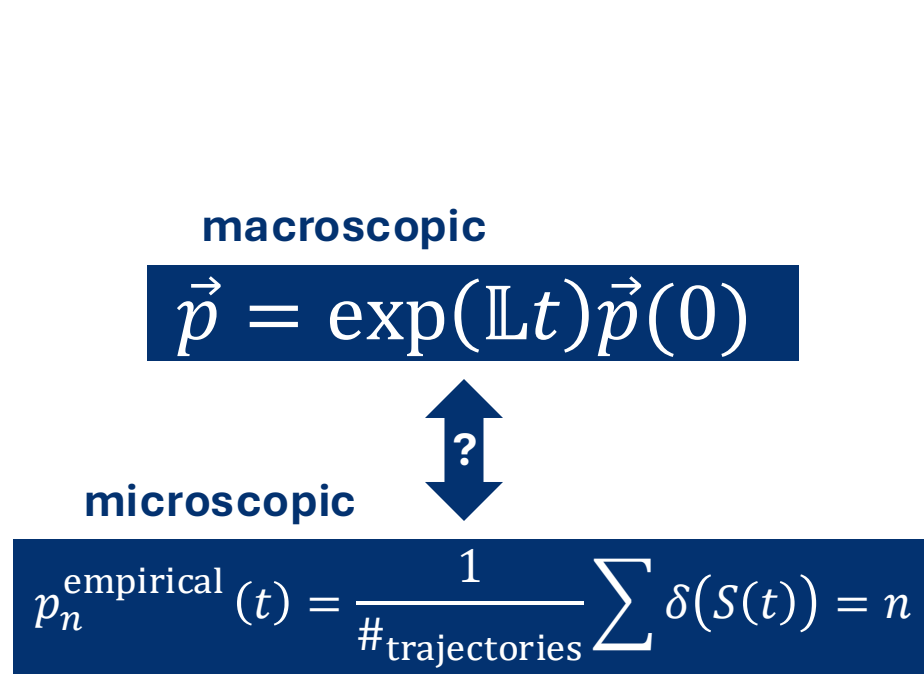
2-State CTMC Microscopic Description



2-State CTMC Microscopic Description



2-State CTMC Microscopic Description



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Compute and Analyze Statistical Quantities

Compute statistical quantities from trajectory data, including **occupation probabilities** and **instantaneous average state currents**.

Equivalence between Macroscopic and Microscopic Descriptions

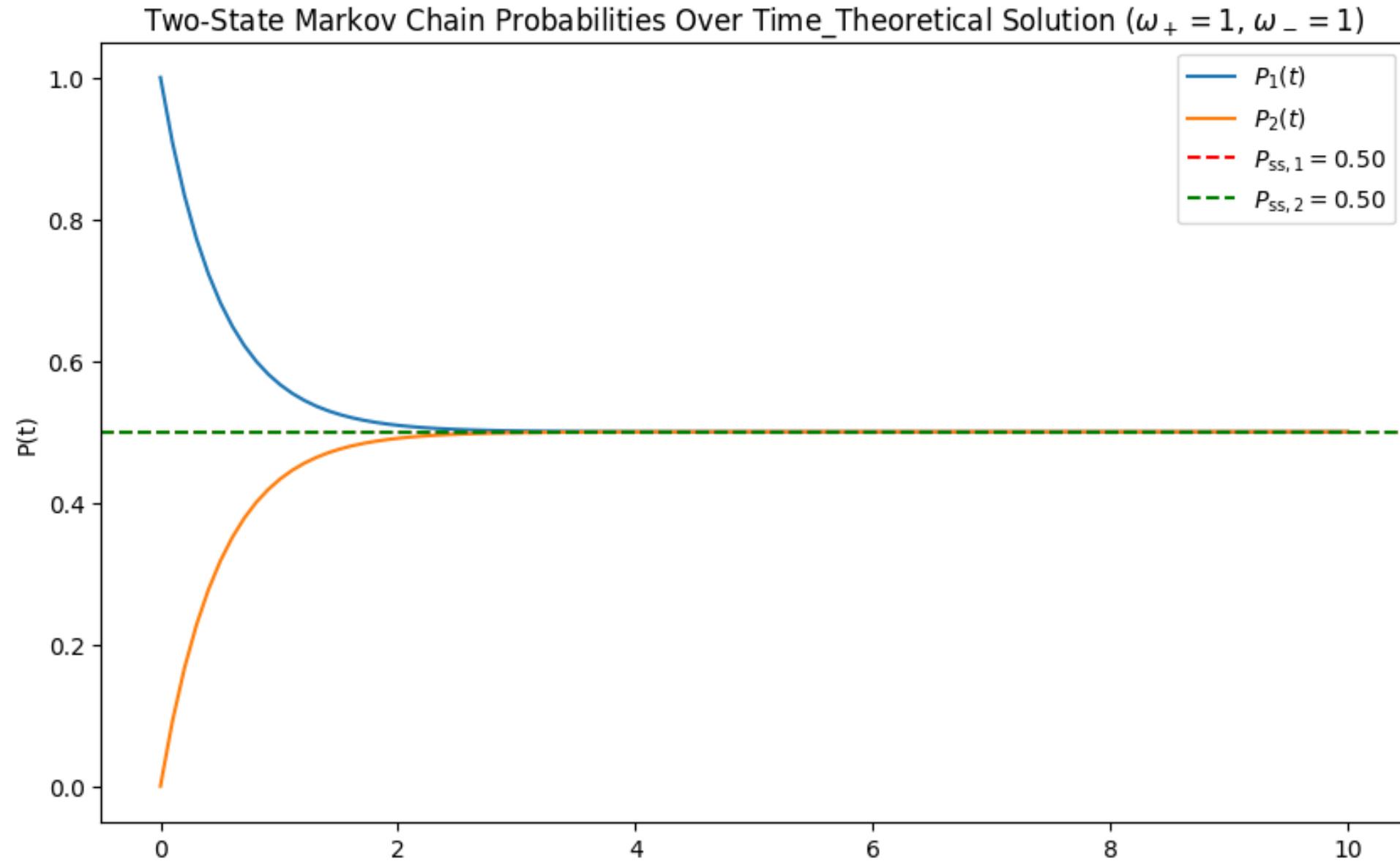
- Macroscopic Description (Theory)

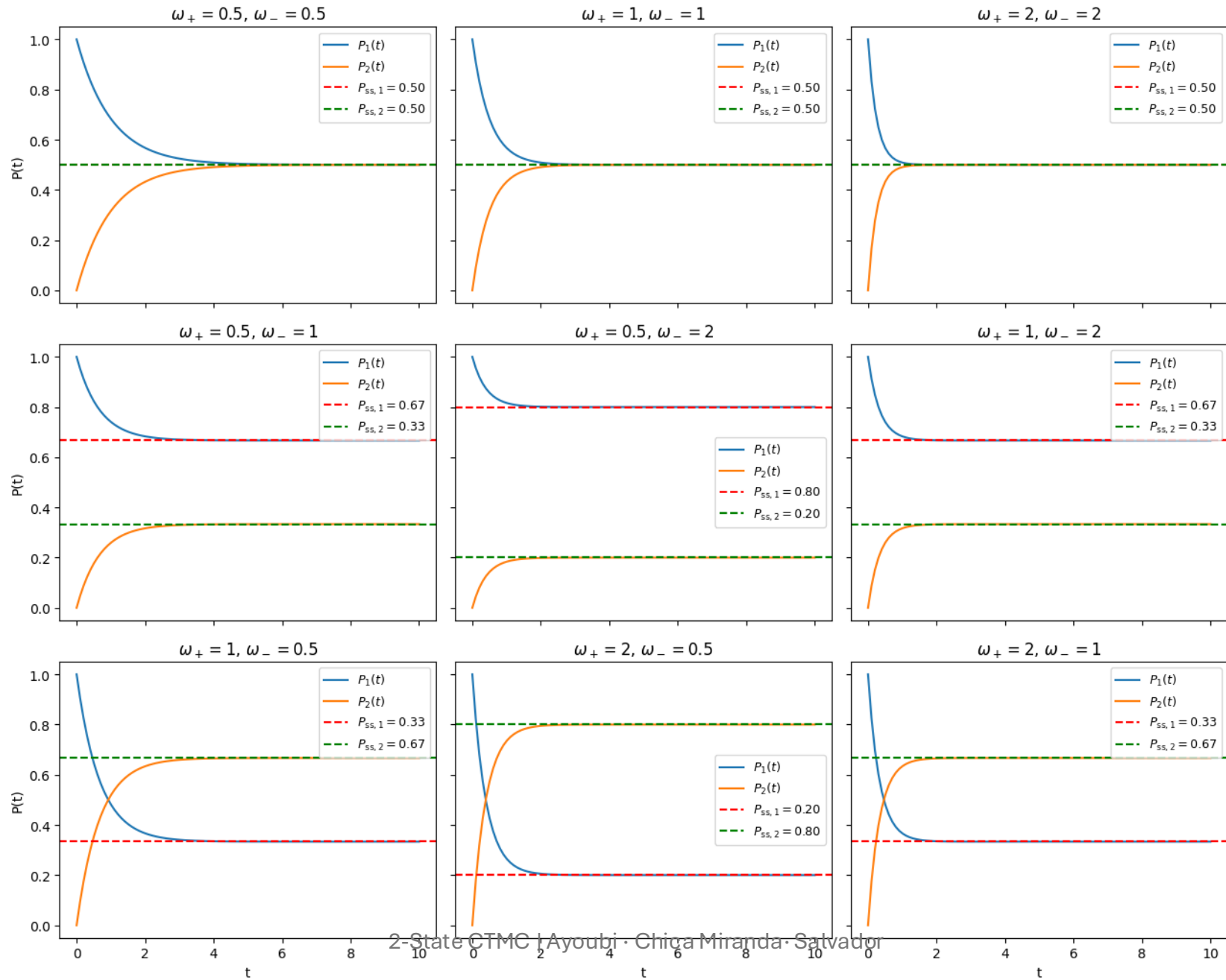
$$\begin{aligned} p_1(t) &= \frac{\omega^-}{\omega^+ + \omega^-} + \frac{\omega^+}{\omega^+ + \omega^-} e^{-(\omega^+ + \omega^-)t}, \\ p_2(t) &= \frac{\omega^+}{\omega^+ + \omega^-} - \frac{\omega^+}{\omega^+ + \omega^-} e^{-(\omega^+ + \omega^-)t}. \end{aligned} \quad p_{1,ss} = \frac{\omega^-}{\omega^+ + \omega^-}, \quad p_{2,ss} = \frac{\omega^+}{\omega^+ + \omega^-}.$$

- Microscopic Description (Simulation)

$$P_1^{empirical}(t) = \frac{1}{\#_{trajs}} \sum_{trajs} \delta(S(t) = 1)$$

Macroscopic Description (Theory)

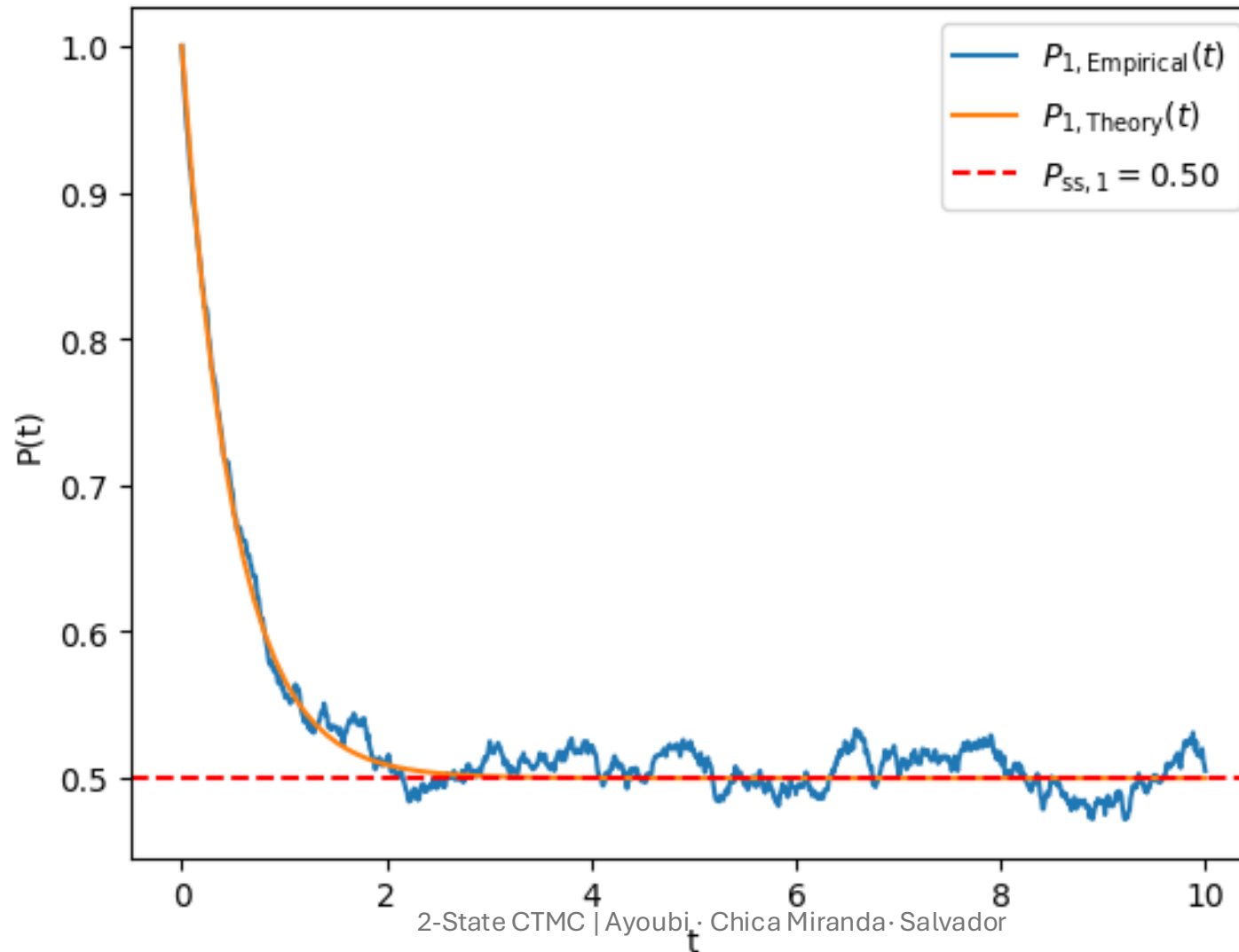




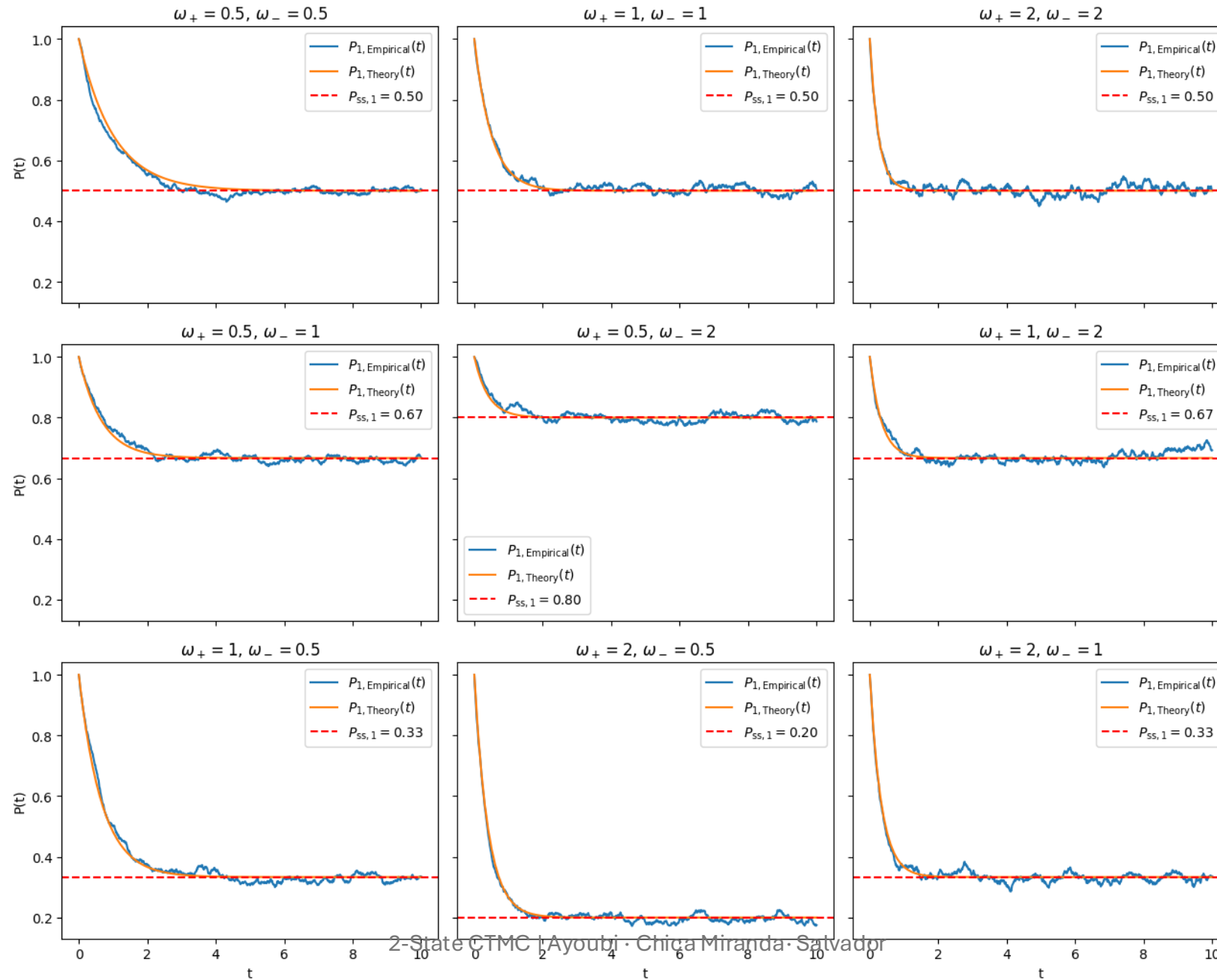
Microscopic Description (Simulation)

Two-State Markov Chain Probabilities Over Time (Theoretical and Empirical)

$$\omega_+ = 1, \omega_- = 1$$



Two-State Markov Chain Probabilities Over Time (Theoretical and Empirical)



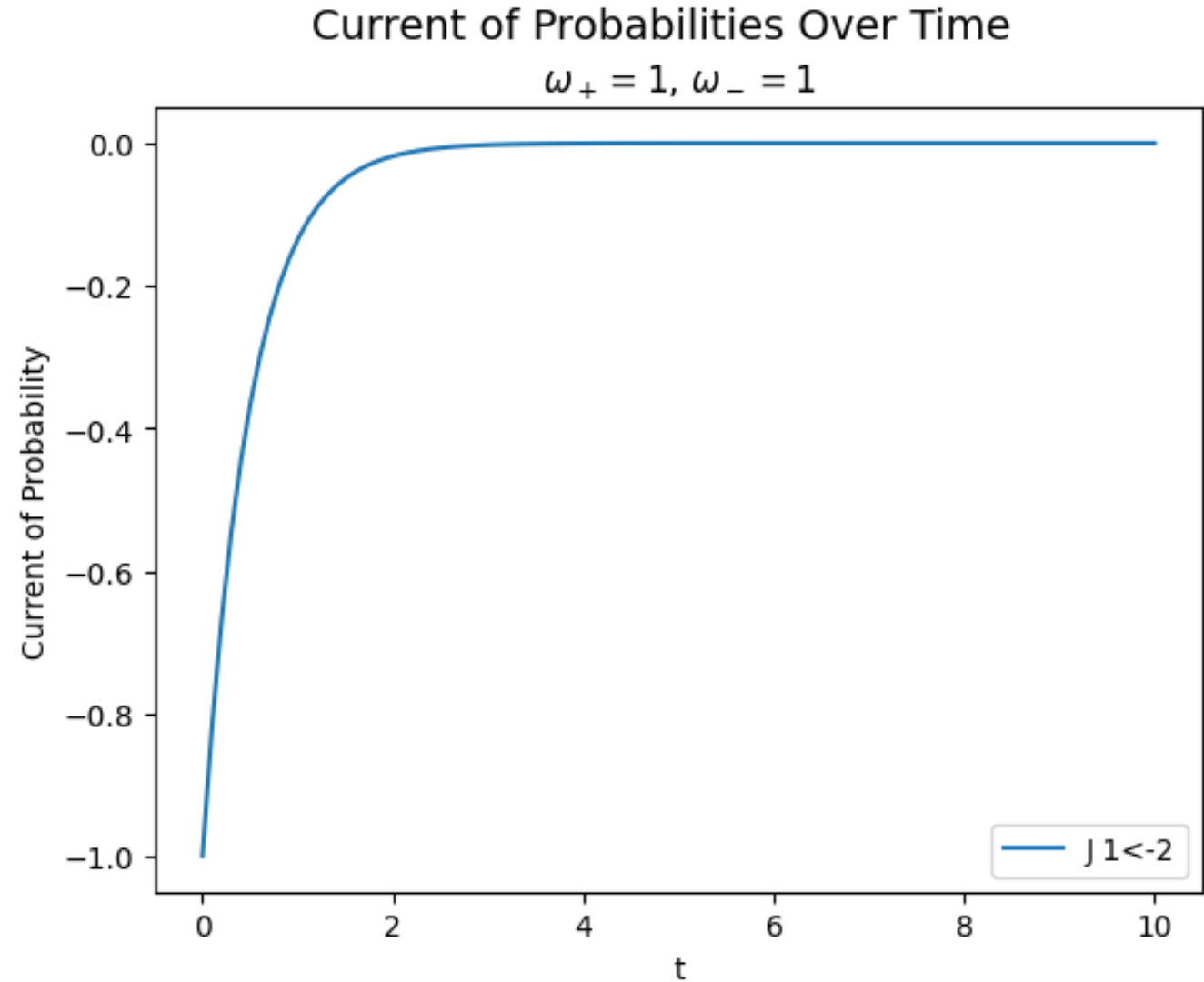
Currents

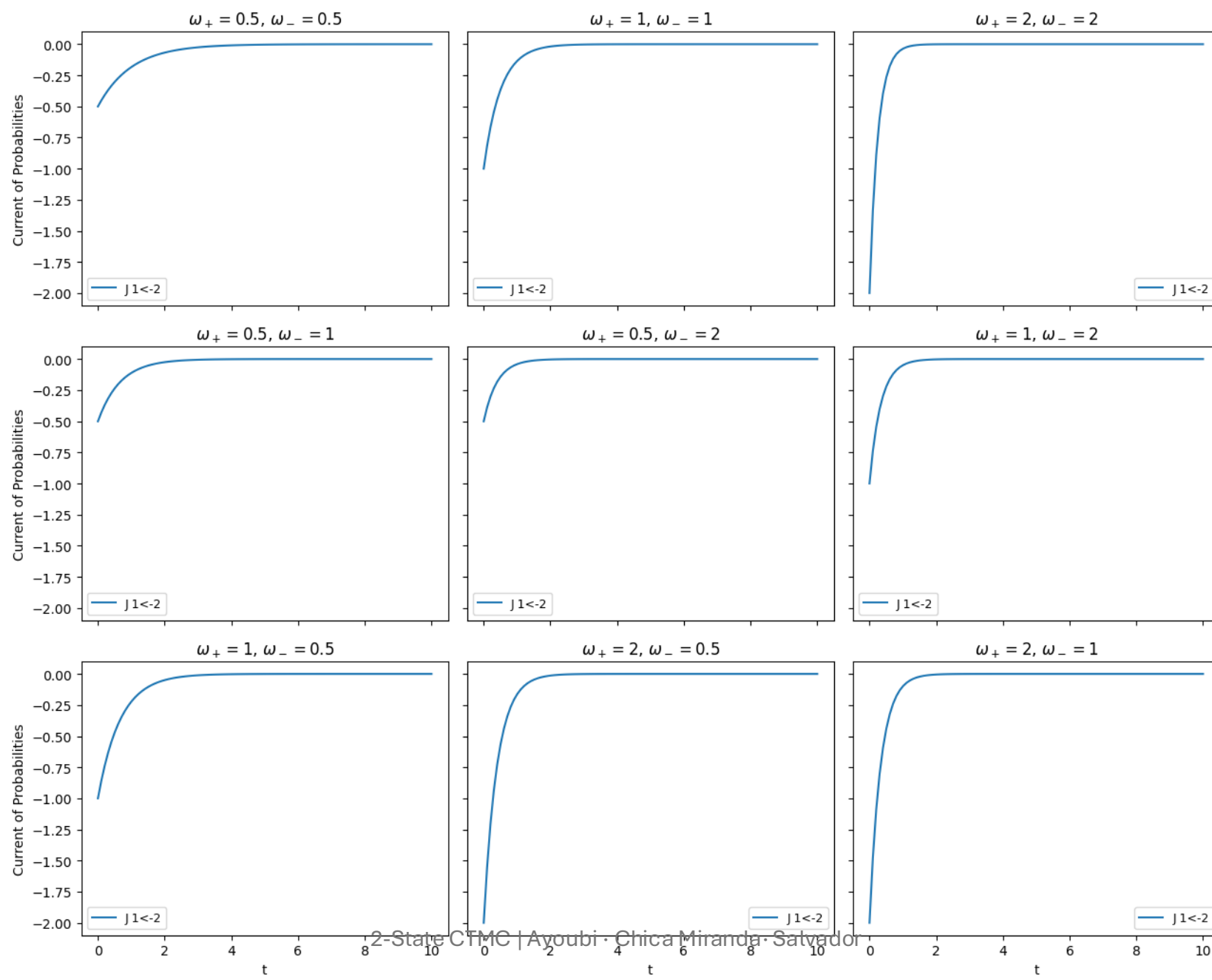
$$\lambda = \omega^+ + \omega^-$$

$$p_1(t) = \frac{\omega^-}{\lambda} + \frac{\omega^+}{\lambda} e^{-\lambda t}$$

$$p_2(t) = \frac{\omega^+}{\lambda} - \frac{\omega^+}{\lambda} e^{-\lambda t}$$

$$\begin{aligned} J_{1 \leftarrow 2}(t) &= \omega^- \left(\frac{\omega^+}{\lambda} - \frac{\omega^+}{\lambda} e^{-\lambda t} \right) - \omega^+ \left(\frac{\omega^-}{\lambda} + \frac{\omega^+}{\lambda} e^{-\lambda t} \right) \\ &= \frac{\omega^+ \omega^-}{\lambda} - \frac{\omega^+ \omega^-}{\lambda} e^{-\lambda t} - \frac{\omega^+ \omega^-}{\lambda} - \frac{\omega^{+2}}{\lambda} e^{-\lambda t} \\ &= -e^{-\lambda t} \frac{\omega^+ (\omega^- + \omega^+)}{\lambda} \\ &= \boxed{-\omega^+ e^{-\lambda t}}. \end{aligned}$$





Markov Processes: From Two-State Systems to N-State Generalizations

1. Introduction to Markov Processes

1.1 Mathematical Foundation

A sequence $(X_t)_{t \in [1, n]}$ is a Markov process if:

$$P \left(x_{t+1} \mid \bigcap_{i=1}^t x_i \right) = P(x_{t+1} \mid x_t)$$

- Memoryless property: Future depends only on current state
- Discrete states $\rightarrow$ Markov chain
- Continuous time $\rightarrow$ Continuous Markov chain

2. Continuous Markov Chains

2.1 Transition Rates

For infinitesimal time Δt :

$$P(X(t + \Delta t) = j \mid X(t) = i) = W_{j \leftarrow i} \Delta t (j \neq i)$$

$$P(X(t + \Delta t) = i \mid X(t) = i) = 1 - \sum_{j \neq i} W_{j \leftarrow i} \Delta t$$

2.2 Master Equation

Using the Law of Total Probability:

$$P_i(t + \Delta t) = P_i(t) + \sum_{j \neq i} [W_{i \leftarrow j} P_j(t) - W_{j \leftarrow i} P_i(t)] \Delta t$$

Taking limit $\Delta t \rightarrow 0$:

$$\frac{dP_i(t)}{dt} = \sum_{j \neq i} [W_{i \leftarrow j} P_j(t) - W_{j \leftarrow i} P_i(t)]$$

3. Matrix Formulation

3.1 Generator Matrix

Vector form: $\frac{d\vec{P}}{dt} = \hat{L}\vec{P}(t)$

Where the generator matrix has elements:

$$L_{i,j} = W_{i \leftarrow j} - \delta_{i,j} \sum_{k=1}^n W_{k \leftarrow i}$$

3.2 Conservation Property

$$L_j = \sum_{i=1}^n L_{i,j} = 0$$

This ensures probability conservation: $\sum_i P_i(t) = 1$

4. Spectral Solution Method

4.1 General Solution

$$\vec{P}(t) = e^{\hat{L}t} P(0)$$

4.2 Eigenvalue Decomposition

$$\vec{P}(t) = \sum_{i=1}^n c_i \vec{v}_{\lambda_i} e^{\lambda_i t}$$

Where:

- λ_i are eigenvalues of $\hat{L}$
- $\vec{v}_{\lambda_i}$ are corresponding eigenvectors
- $c_i = \hat{B}^{-1} \vec{P}(0)$ from initial conditions

4.3 Key Spectral Properties

- At least one eigenvalue $\lambda = 0 \rightarrow$ Stationary state
- All other eigenvalues have $\lambda < 0 \rightarrow$ Decaying modes
- As $t \rightarrow \infty$: $\vec{P}(t) \rightarrow c_0 \vec{v}_0$ (steady state)

5. Probability Currents

5.1 Current Definition

Probability current from state j to state i :

$$J_{i \leftarrow j}(t) = W_{i \leftarrow j} P_j(t) - W_{j \leftarrow i} P_i(t)$$

5.2 Master Equation in Current Form

$$\frac{dP_i(t)}{dt} = \sum_{j \neq i} J_{i \leftarrow j}(t)$$

5.3 Equilibrium Condition

In equilibrium: No net probability currents (except possible cyclic steady-state currents)

6. Central Limit Theorem for Markov Processes

6.1 Sum of Bernoulli trials

X_i is a Bernoulli random variable with success probability p .

$$X_i = \begin{cases} 1 & \text{wp } p \\ 0 & \text{wp } 1 - p \end{cases}$$

The expected value $\mathbb{E}$ and variance $\mathbb{V}$ are:

$$\mathbb{E}[X_i] = p$$

$$\mathbb{V}[X_i] = p(1 - p)$$

S_n is the sum of n i.i.d. Bernoulli variables (a Binomial count).

$$S_n = \sum_{i=1}^n X_i$$

By the Central Limit Theorem (CLT), the standardized sum Y_n converges to the standard normal distribution $\mathcal{N}(0, 1)$:

$$S_n = a_n + b_n Y_n$$

where:

$$a_n = \mathbb{E}[S_n] = np$$

$$b_n = \sqrt{\mathbb{V}[S_n]} = \sqrt{np(1 - p)}$$

And the convergence in distribution is:

$$Y_n = \frac{S_n - a_n}{b_n} \rightarrow \mathcal{N}(0, 1)$$

6.2 Sum of Sample Proportions ($\mathcal{S}_m$)

$p_n^{(i)}$ is the sample proportion (sample mean) from the i^{th} block of n trials. The variables $p_n^{(i)}$ are i.i.d.

$$p_n^{(i)} = \frac{1}{n} S_n$$

Moments of the Sample Proportion:

$$\mathbb{E}[p_n^{(i)}] = \mathbb{E}\left[\frac{1}{n} S_n\right] = p$$

$$\mathbb{V}[p_n^{(i)}] = \mathbb{V}\left[\frac{1}{n} S_n\right] = \frac{1}{n^2} \mathbb{V}[S_n] = \frac{p(1 - p)}{n}$$

$\mathcal{S}_m$ is the sum of m sample proportions. Applying the CLT to this sum:

$$\mathcal{S}_m = \sum_{i=1}^m p_n^{(i)} = A_m + B_m Y'_m$$

where:

$$A_m = \mathbb{E}[\mathcal{S}_m] = m \cdot p$$

$$B_m = \sqrt{m \cdot \mathbb{V}[p_n^{(i)}]} = \sqrt{\frac{p(1 - p)m}{n}}$$

The standardized sum Y'_m converges to the standard normal in distribution:

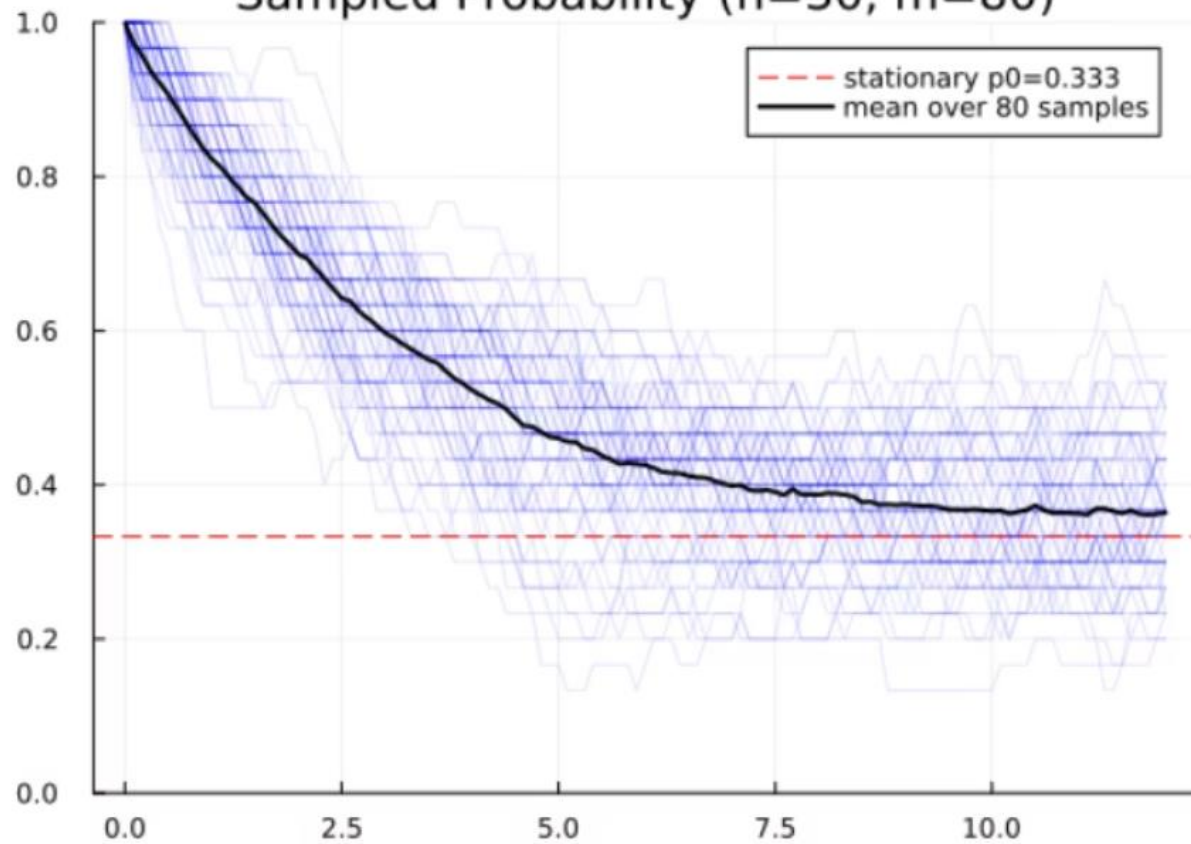
6.3 Confidence Intervals

95% confidence interval for probability estimates:

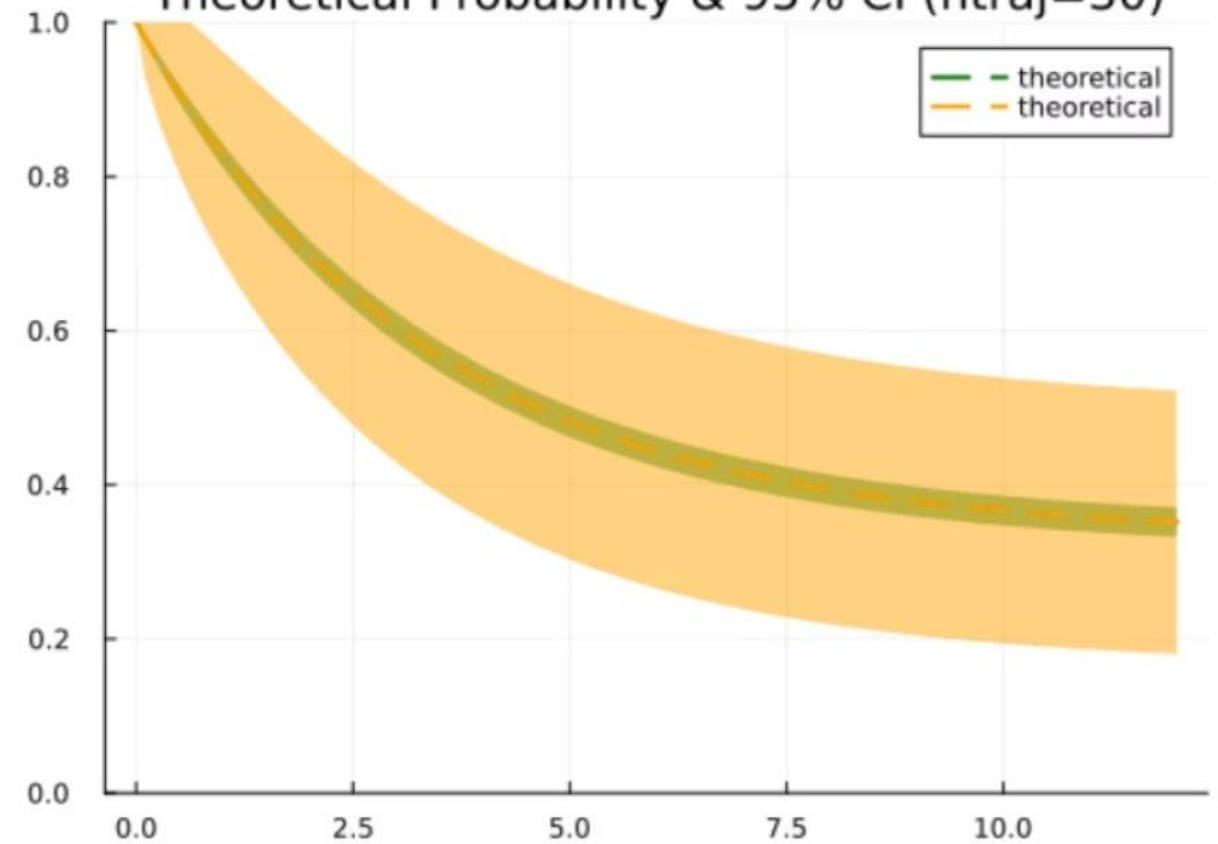
$$CI = p \pm 1.96 \cdot \sqrt{p(1-p)/n \cdot m}$$

[

Sampled Probability (n=30, m=80)



Theoretical Probability & 95% CI (ntraj=30)



7. Multi-State Applications

7.1 Cyclic Systems

State transitions: $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$

Features:

- Can have equal or different transition rates
- May exhibit persistent probability currents in steady state

7.3 Weakly Coupled Systems

Structure: Strong connections between some states, weak connections to others

Features

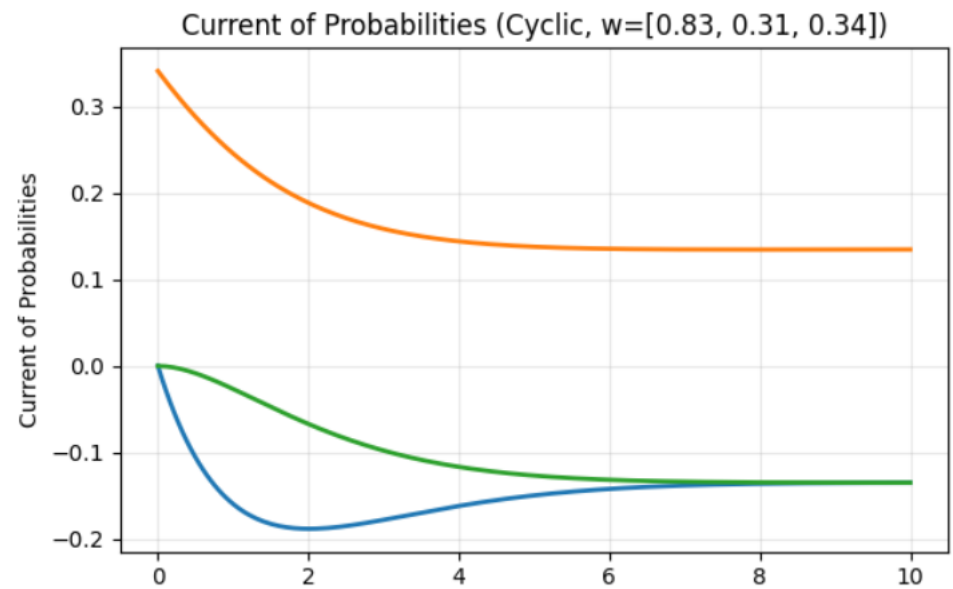
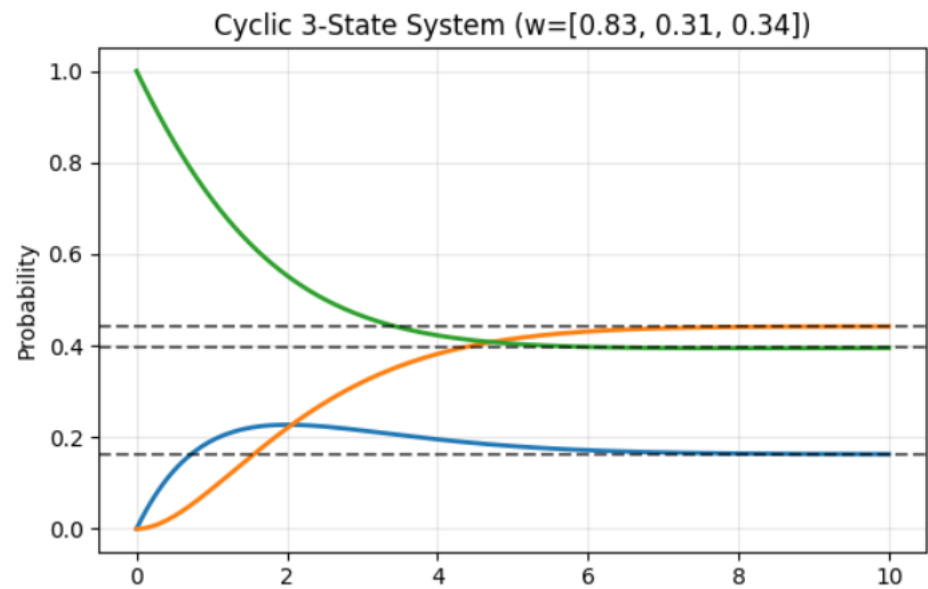
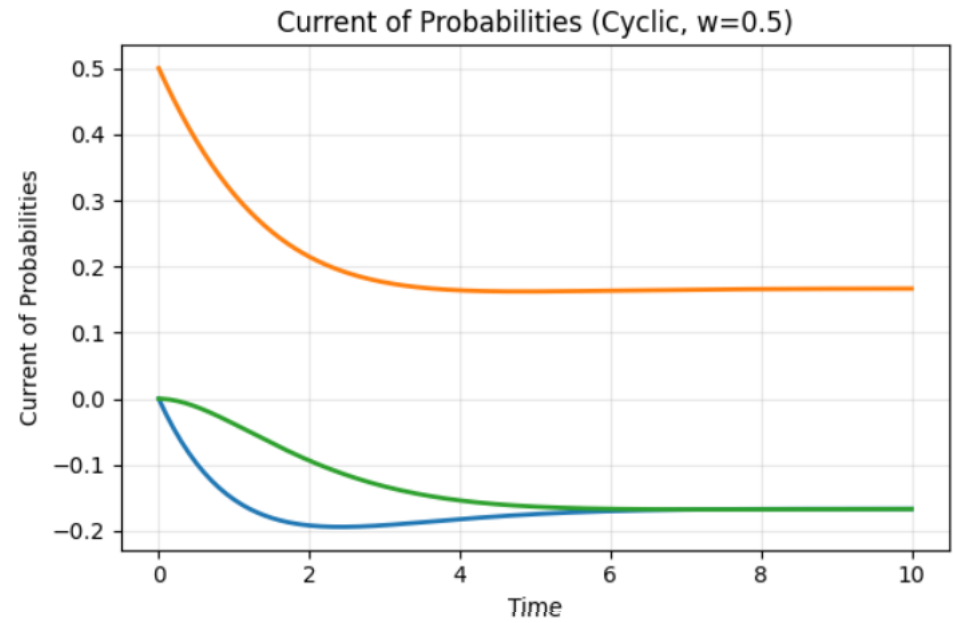
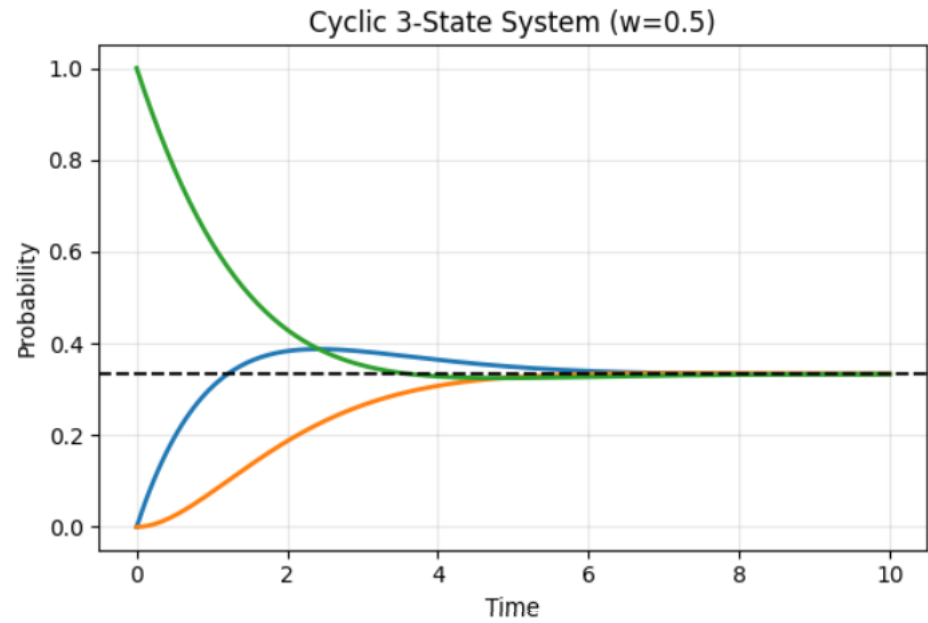
- Fast vs slow relaxation modes
- Cluster formation in probability space
- Multiple time scale dynamics

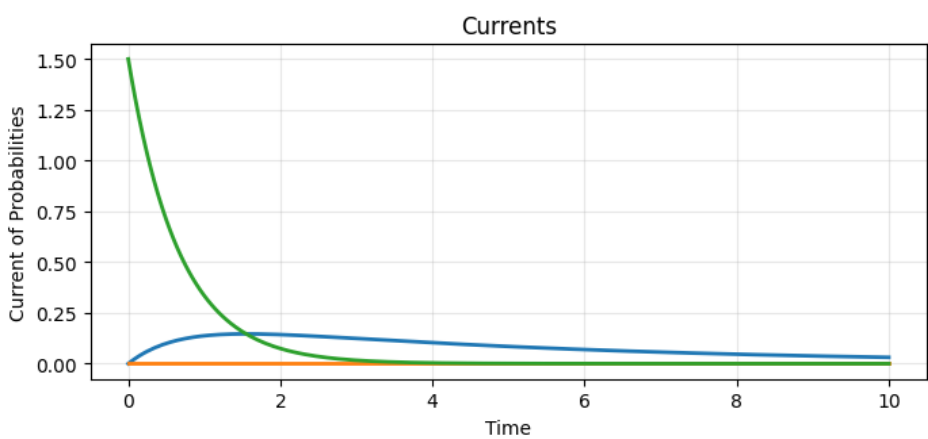
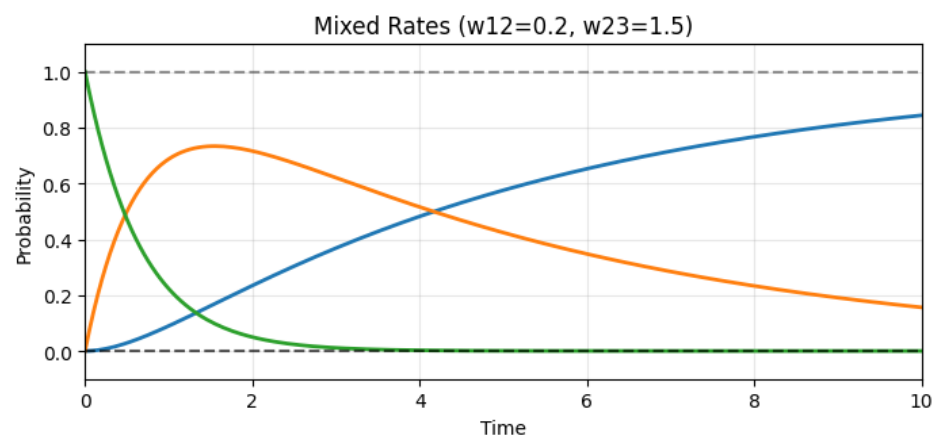
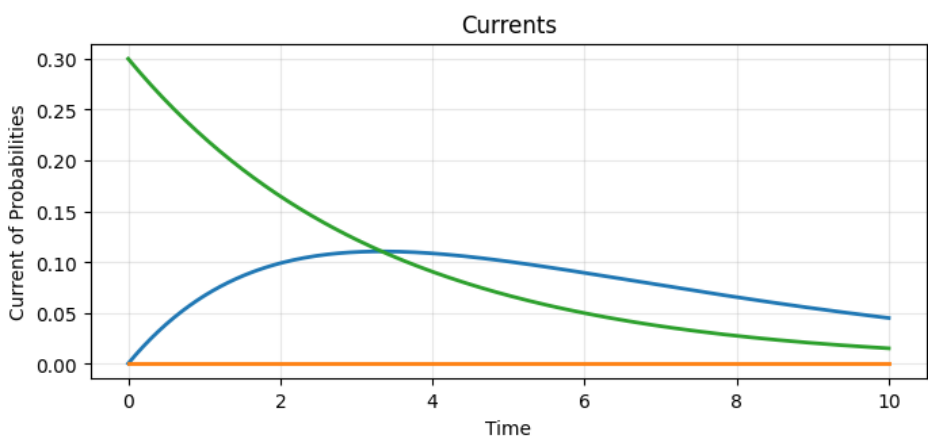
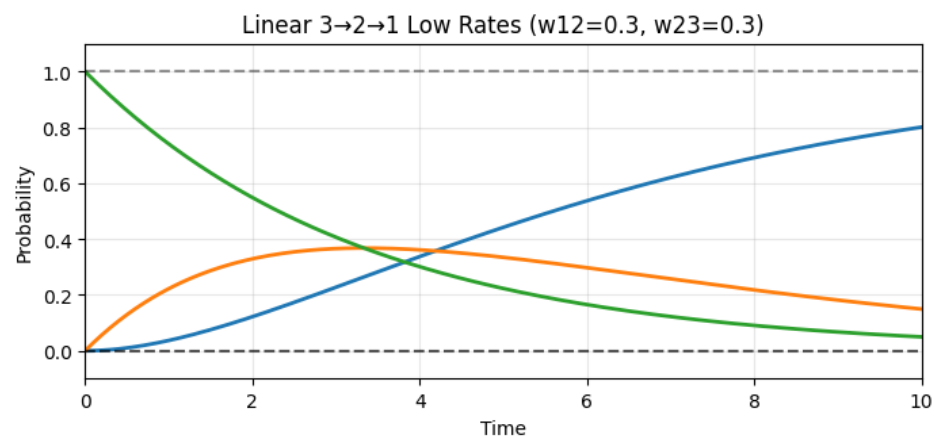
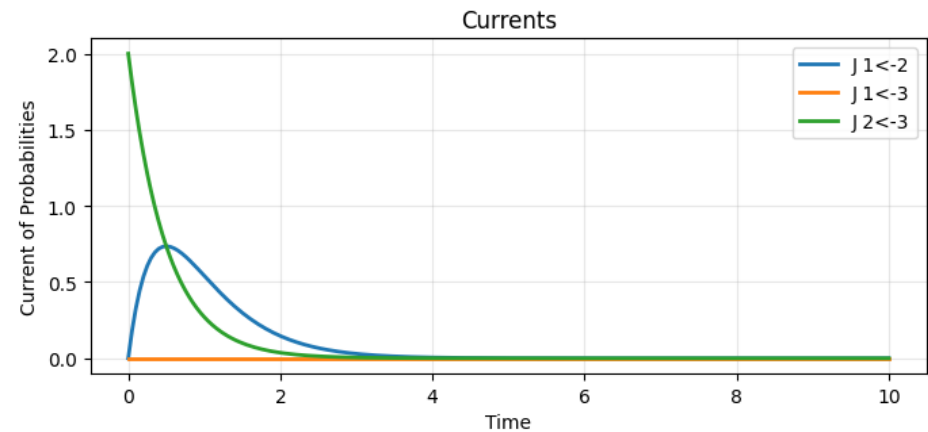
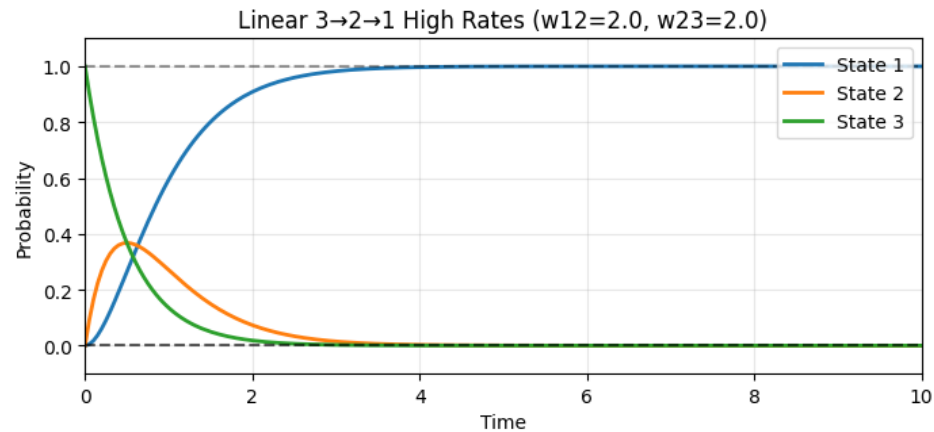
7.2 Linear Cascade Systems

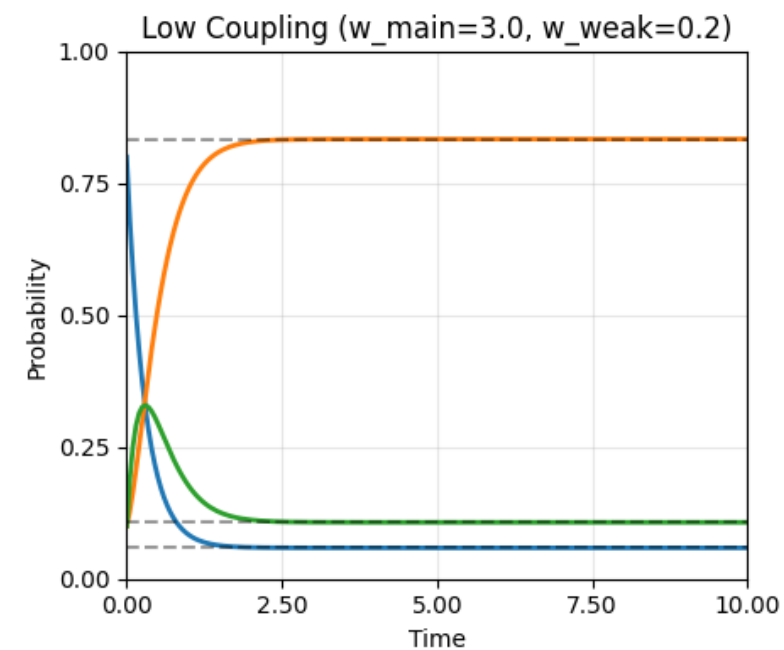
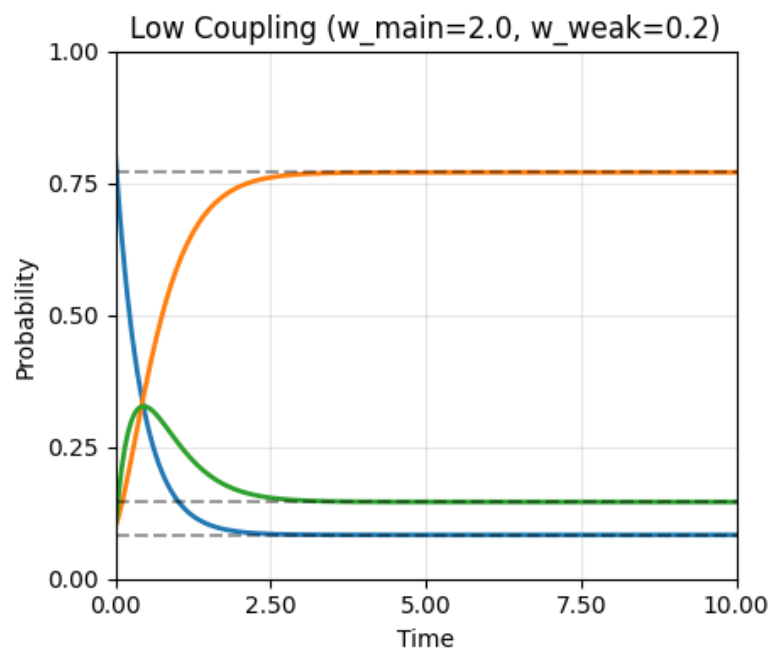
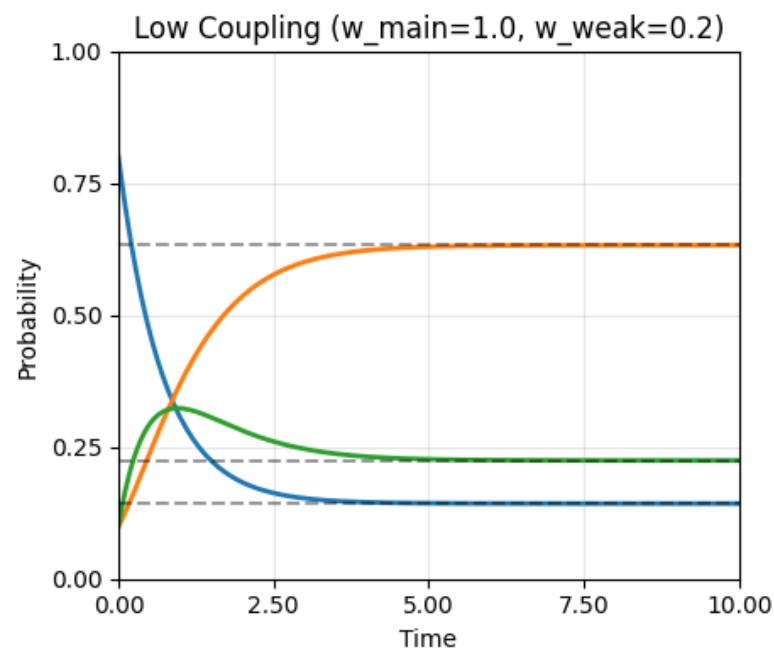
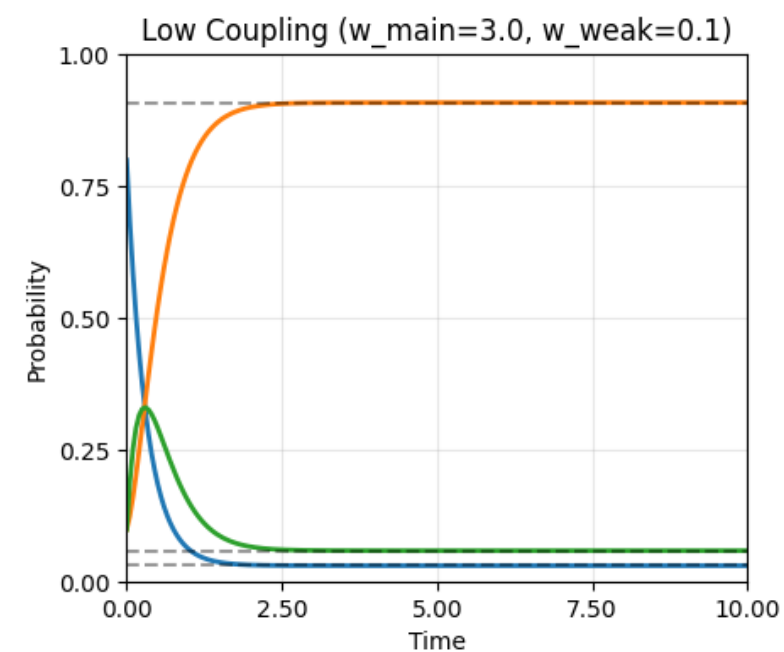
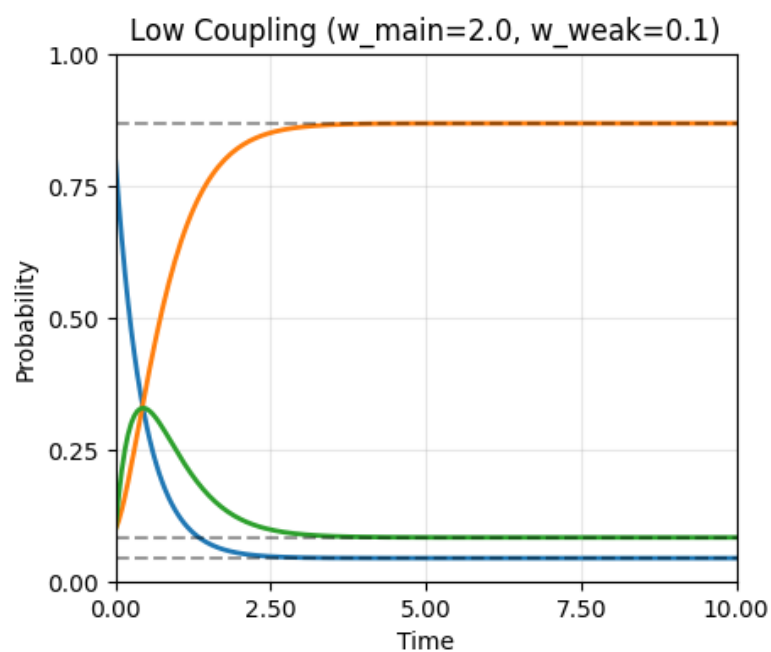
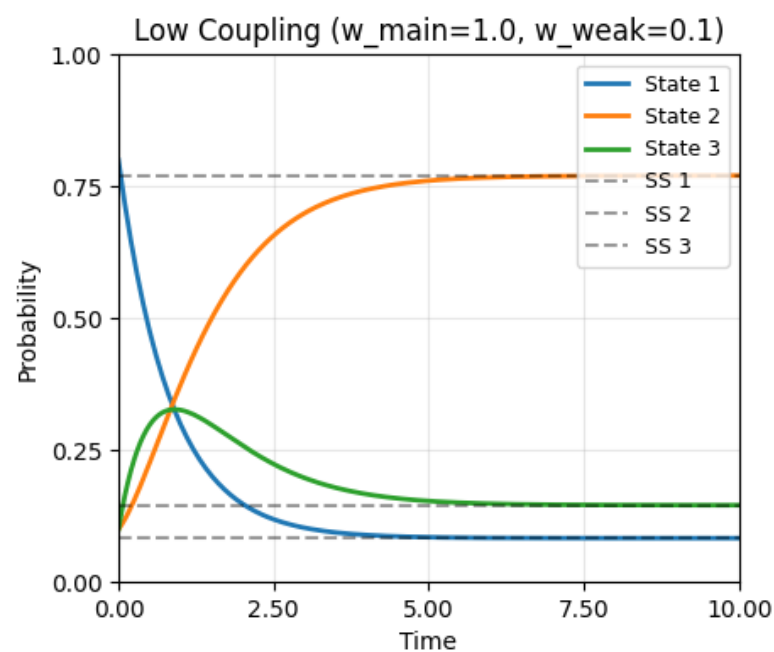
State transitions: $3 \rightarrow 2 \rightarrow 1$

Features:

- Unidirectional flow
- Eventual absorption in terminal states









8. Methodological Considerations

8.1 When to Use Spectral Methods

- Low-dimensional systems (small n)
- Dense matrices
- Exact analytical solutions needed
- Theoretical analysis

8.2 Alternative Approaches

- Numerical ODE solvers: For moderate-sized systems
- Matrix exponentiation: For small to medium systems
- Gillespie algorithm: For stochastic simulations
- Sparse matrix methods: For high-dimensional systems

9. Conclusions

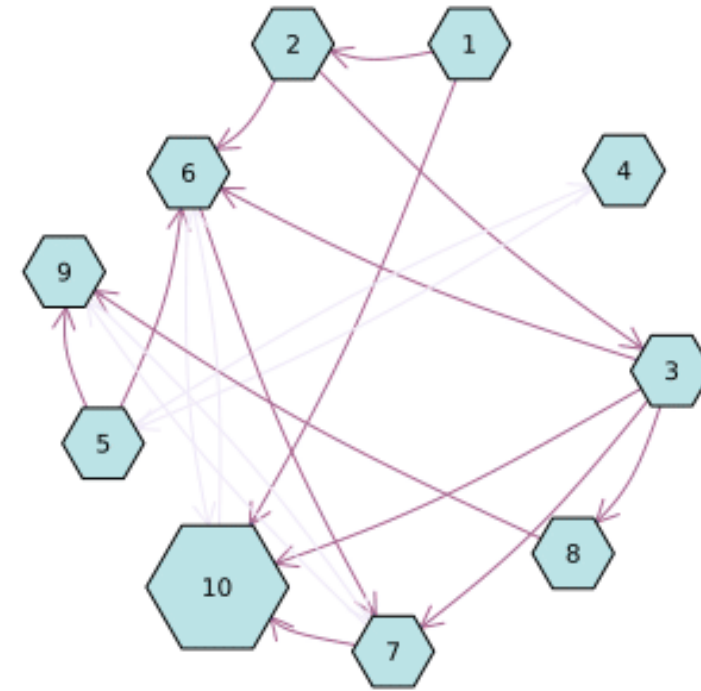
9.1 Key Insights

- Markov formalism provides powerful framework for multi-state systems
- Spectral decomposition offers analytical solutions for low-dimensional cases
- Probability currents reveal detailed dynamical behavior
- System topology (cyclic, linear, coupled) determines qualitative behavior
- Statistical analysis enables confidence estimation from finite trajectories

9.2 Broader Implications

- Universal mathematical structure across diverse applications
- Flexible framework for modeling complex dynamical systems
- Bridge between discrete and continuous descriptions
- Foundation for stochastic and non-equilibrium thermodynamics
- Robust statistical inference from limited experimental data

Time: 0.0



References

- **Marsili, M.** (2025). *Lecture on Probability, Information and Large Scale Behaviour* (Chapter 11: Markov Chains). ISBN: 9788898587063 (electronic edition). DOI: 10.22323/9788898587063.
Keywords: Probability, Information theory, Laws of large numbers, Limit theorems, Large deviations, Statistical mechanics, Statistical inference.
- **Roldán, É., Lisica, A., Sánchez-Taltavull, D., & Grill, S. W.** (2016). *Stochastic resetting in backtrack recovery by RNA polymerases*. *Physical Review E*, 93(6), 062411. DOI: 10.1103/PhysRevE.93.062411.
- **Song, Y. M.** (2025). *Stochastic Simulation Algorithms for Biochemical Reaction Networks*. Lecture slides, *International Workshop on Mathematical Biology 2025*.